BLADE-NCO Theory Research
Stage 1 with Stage 1.5 Formal-Model Corrections

Research note

October 21, 2026

Abstract

This note formalizes structural reachability, inherited population memory, moving hard-instance regions, and first discovery measured in computational cost. It records three candidate claims, attempts a proof of a finite-window memory comparison, and tests broad memory/diversity claims against exact finite-state examples. The strongest surviving result is conditional: retained structural memory helps only when it supplies a verified per-cost hit-hazard advantage over the comparator. No unconditional BLADE-NCO advantage is established. This is a Stage 1 note, not a completed theory paper.

Stage 1.5 corrects the state space, time indices, probability space, aggregation language, and implementation mapping without adding a theorem or changing that conclusion. The finite examples remain counterexamples to universal claims; they are not assumptions that the TSP instance space is finite.

1 Scope and status

The scope is restricted to structural reachability, evolutionary memory, and expected first-hitting time under a non-stationary hardness landscape. The supplied BLADE-NCO context is FOGA-11 structural mutation, inherited parent populations, HGB40 hardness estimation, Elite-QD1000 (P=T), and BQ solver co-adaptation. Those names are treated as labels, not as complete algorithm specifications.

Statuses are reported on three separate axes. [**Mathematical status: PROVED**], for example, concerns the validity of a mathematical statement under its premises; [**Implementation applicability: VERIFIED, PARTIAL, OPEN, or NOT APPLICABLE**] concerns whether those premises and variables match the inspected S6-AE implementation; and [**Scientific novelty: OPEN or NOT CLAIMED**] concerns literature priority. A proof of a conditional probability bound does not establish implementation applicability or novelty. Compilation verifies only that the source typesets; it does not verify mathematical correctness.

The Stage 1.5 implementation boundary is the verified HGB40 + Elite-QD1000 ($P = T$) S6-AE path. Its frozen artifacts passed 40/40 strict checks. Historical GP4 output is not used as the definition of the current method. Details that could not be established from the inspected code and artifacts are marked OPEN rather than inferred.

2 Prior-theory boundary

Related theory includes rigorous results in which populations track moving optima and in which diversity mechanisms have different effects, but those works analyze explicit dynamic-function classes and specified algorithm dynamics [1, 2]. They do not establish compute-normalized discovery of solver-relative hard instances under solver co-adaptation. Thus a generic claim that "memory/diversity helps in dynamic optimization" is unsuitable as a novelty claim. The present abstraction differs in its target: discovering instances whose measured hardness changes as the solver co-adapts, under a compute budget and with a surrogate-based archive. Whether that narrower combination is novel remains **[Scientific novelty: OPEN]**; this note is not a literature-priority search.

3 Formal model and implementation boundary

3.1 Structural states and mutation

Let $\mathcal{Z}$ be the set of encoded candidate instances and let $\phi : \mathcal{Z} \to \mathcal{X}$ be a measurable map from encodings to the solver-visible TSP instance. For the inspected fixed-size experiments one may take $\mathcal{X}$ to be the valid canonical subset of $[0,1]^{100\times 2}$, equipped with its Borel σ-algebra $\mathcal{B}_{\mathcal{X}}$; a variable-size model may instead use a measurable disjoint union over instance sizes. Encodings z, z' are equivalent when $\phi(z) = \phi(z')$. A structural state is a phenotype $x \in \mathcal{X}$, for example a canonical graph/instance together with whichever structural attributes determine solver input. The exact canonicalization and attributes are implementation questions. We assume ϕ and every event used below are measurable.

The instance space is not made finite by a finite search budget. A realized finite-budget run contains finitely many attempted instances, but the support of a continuous-coordinate mutation kernel can remain uncountable. For each BQ cycle k and FOGA operator o, let $K_{k,o}(x, A \mid s)$ be a Markov kernel from a parent x and full pre-mutation state s into measurable $A \in \mathcal{B}_{\mathcal{X}}$. It includes stochastic operator parameters, repair, and canonicalization. Dependence on k or s permits solver-, archive-, or history-dependent behavior. The corresponding encoding process need not remain Markov after projection through ϕ.

Let $d : \mathcal{X} \to \mathbb{R}^4$ be the measurable QD descriptor map, including the frozen normalization, and let $g : \mathcal{X} \to \{1, \dots, K\}$ be nearest-centroid assignment with a declared measurable tie rule. Then $\mathcal{N}_j = g^{-1}(\{j\})$ gives a finite measurable partition after discarding any empty cells. Finiteness belongs to the niche index set, not to $\mathcal{X}$.

3.2 Two time scales and the S6-AE batch

Use k only for the BQ adaptation cycle. At the start of cycle k, the BQ solver state q_k and the parent population $P_k = (X_{k,1}, \dots, X_{k,m})$ are fixed, with $m = 10{,}000$ in S6-AE. Let $\mathcal{O}$ be the ordered list of the 11 FOGA operators. The within-cycle mutation attempts are indexed by $(i, o) \in \{1, \dots, m\} \times \mathcal{O}$:

$$Z_{k,i,o} \sim K_{k,o}(X_{k,i}, \cdot \mid S_k).$$

This notation denotes a conditional law, not random parent or operator selection. The inspected implementation uses the deterministic Cartesian schedule "parents in stored order, then operators in registry order," so each parent–operator pair is attempted exactly once. A content-derived seed from $(k, \mathrm{hash}(X_{k,i}), o)$ drives the stochastic operator parameters and repair. Invalid and duplicate outputs are recorded and are not separate resampling opportunities.

Let O_k be the accepted, deduplicated children and let $C_k = \mathrm{unique}(P_k \cup O_k)$ be the candidate union, retaining the first occurrence under the declared order. At C00 S6-AE reuses the hash-locked corrected union; later cycles construct this union directly. The cycle then fits/scores with HGB40, selects the acquisition/next-parent set, obtains exact labels for that selected set, and only then adapts the BQ solver from q_k to q_{k+1}. Thus within-cycle mutation attempts and solver cycles must not share one index.

3.3 Population memory and parent survival

At BQ cycle k, let P_k be the ordered parent population and A_k the branch-local labeled archive. Let S_k collect scores, lineage information, solver state, and other algorithm variables that affect the next transition. Thus $M_k = (P_k, A_k, S_k)$ is the operational memory state. Parent survival is a transition event: for parent i, its survival probability is

$$r_{k,i} = \mathbb{P}(X_{k,i} \in P_{k+1} \mid \mathcal{F}_k^{\mathrm{pre}}).$$

In a generic policy this may differ from a proposal-selection probability. For S6-AE, however, no such random parent-selection variable is present: every member of P_k receives all 11 scheduled mutations. Population update is a deterministic function of the realized candidate union, HGB40 scores, niche assignments, and hash tie-breaks. Randomness can still enter through the roots, mutation parameters, calibration sample, model fitting, training permutations, or nondeterministic numerical execution. The full process may be time-inhomogeneous Markovian in a sufficiently rich (M_k, θ_k, q_k); a niche label or count vector alone need not be.

In verified S6-AE, one maximum-current-HGB candidate is retained from each occupied niche, and the remaining slots are filled by global descending HGB score, with ascending instance hash breaking ties. The result has 10,000 unique members and is used both as the BQ acquisition set T_k and next parent population P_{k+1}: $P_{k+1} = T_k$. "Elite-QD1000" refers to at most one representative in each of the $K = 1000$ measurable niche cells; it does not make the instance space finite.

3.4 Probability space and frozen executions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry exactly the random inputs declared for the analysis: possibly random initial roots, content-derived operator seeds and their stochastic parameters/repair, seeded archive calibration sampling, HGB estimator seeds, BQ training permutations and suffix sampling, and any explicit selection randomness in a comparator. Conditioning on the frozen S6-AE roots, source artifacts, software/runtime, and all recorded seeds gives a single realized trajectory ω_0; many nominal "probabilities" then collapse to zero or one. A probabilistic reachability theorem must therefore state which inputs vary across Ω. Repeatedly serializing a deterministic batch does not create random parent/operator selection, and unspecified hardware nondeterminism is not silently treated as a scientific sampling law.

3.5 Hardness and moving target

Let θ_k describe the task/environment and q_k the state of the co-adapting BQ solver before cycle-k adaptation. Define true BQ hardness $h_k^{\mathrm{BQ}}(x) = h(x; \theta_k, q_k)$, where a larger value means harder under a prespecified solver-performance metric. For threshold κ, the true hard region at cycle k is

$$\mathcal{H}_k = \{x \in \mathcal{X} : h_k^{\mathrm{BQ}}(x) \geq \kappa\}.$$

This definition allows the target to move because either the task or solver state changes. During the S6-AE mutation/acquisition part of cycle k, q_k is fixed; adaptation produces q_{k+1} only after T_k is frozen and labeled. For the proofs, the target attached to a declared opportunity must be predictable from its pre-opportunity history.

HGB40 supplies a cycle-specific prediction $\widehat{h}_k(x)$ (in the implementation, predicted $\log(1 + \mathrm{LBQ})$), not the definition of h_k^{BQ} or $\mathcal{H}_k$. Define the latent generation-level event

$$D_k^{\mathrm{gen}} = \{\exists x \in C_k : h_k^{\mathrm{BQ}}(x) \geq \kappa\},$$

and the acquisition-level event

$$D_k^{\mathrm{acq}} = \{\exists x \in T_k : h_k^{\mathrm{BQ}}(x) \geq \kappa\}.$$

The former concerns what the generation-level candidate union contains, including retained parents and accepted children, even when exact hardness was not measured; the latter concerns what survived HGB/QD acquisition and can be verified. They are distinct from the event $\{\widehat{h}_k(x) \geq \widehat{\kappa}\}$ and from post-adaptation hardness $h_{k+1}^{\mathrm{BQ}}(x)$. Without a calibration/ranking-and-retention condition, none of these quantities can be substituted for another.

3.6 First hit and computational normalization

For the abstract survival lemma, index declared comparison opportunities by j; j is not automatically a BQ cycle. An opportunity may be one serialized mutation attempt, one full generation event D_k^{gen}, or one acquisition event D_k^{acq}, but the choice must be fixed before the comparison. Let

$$\tau = \inf\{j \geq 1 : Y_j = 1\}, \qquad C_\tau = \sum_{j=1}^{\tau} c_j,$$

where Y_j is the declared true-hardness discovery event and $c_j > 0$ is the normalized compute cost through resolving opportunity j. The budget-B discovery event is $\{C_\tau \leq B\}$. The cost must include the work that differs between methods (structural generation, HGB40 inference, archive operations, and BQ evaluation/verification, as applicable). Counting generations alone is not computational normalization. The elementary bound below uses deterministic opportunity costs c_i; random or method-dependent costs require a cost-aware hazard analysis and are not silently replaced by generation counts.

For a policy π, define its conditional hit hazard

$$q_j^\pi = \mathbb{P}(Y_j = 1 \mid \mathcal{F}_{j-1}^\pi),$$

where $\mathcal{F}_{j-1}^\pi$ is the complete pre-opportunity history. On a no-hit history, a lower bound on q_j^π is a reachability certificate. A Markov state representation is one way to calculate this hazard, but the history-based definition does not require Markovianity.

3.7 Abstraction versus verified BLADE-NCO context

Supplied component	Representation here	Unspecified for a BLADE-specific proof
FOGA-11 mutation	Eleven kernels $K_{k,o}$ on $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$	Exact kernel laws over stochastic parameters and support bounds
Inherited parents; Elite-QD1000 (P=T)	P_k, measurable partition g, deterministic HGB/QD selection, $P_{k+1} = T_k$	Distributional survival/reachability bounds under a declared ensemble
HGB40 and BQ co-adaptation	Prediction $\widehat{h}_k$, true BQ hardness h_k^{BQ}, solver state q_k	Scientific threshold, calibration/retention bridge, and matched validation cost
Fixed generator and HAC	Comparator proposal kernels	Exact proposal laws, gradient access, stopping rules, and per-opportunity compute
Finite budget	Cumulative cost C_τ	Which compute resources are charged and how variable costs are normalized

Accordingly, the formal model is a deliberately explicit abstraction. The verified schedule and selection mapping do not assert unobserved transition probabilities. The remaining implementation-to-theory gaps are recorded as open assumptions in `assumption_register.md`.

3.8 Niche aggregation and the Markov condition

Suppose a Markov kernel Q on a full state space is aggregated by a finite measurable partition $\{B_1, \ldots, B_L\}$. The strong-lumpability equalities

$$Q(s, B_j) = Q(s', B_j) \quad \text{for every } s, s' \in B_i \text{ and every } i, j$$

are sufficient for the aggregate block process to be Markov for every initial law. They are not necessary for a particular initial law. A weaker, distribution-dependent Markov aggregation may hold when the conditional within-block law induced by that initial distribution and history makes the averaged transition to each B_j history-independent (possibly with a time-dependent aggregate kernel). Such a claim must name that law and prove the conditional equality; it is not obtained merely by averaging once.

The counterexample in Section 6.1 shows failure of the strong condition and therefore invalidates an implementation-independent Markov claim based on niche alone. It does not rule out a special weak aggregation for a specified initial distribution. In S6-AE, niche summaries hide parent identity, within-niche HGB order, candidate age, solver/archive state, and the deterministic hash tie-break, so strong lumpability is not established and a distribution-dependent alternative remains OPEN.

4 Candidate propositions

4.1 Candidate 1: finite-window reachability comparison

[**Mathematical status: CONDITIONAL**] [**Implementation applicability: PARTIAL**] [**Scientific novelty: NOT CLAIMED**] The following is the strongest surviving candidate.

Proposition 1 (Memory advantage on a stable target window)**.** *Suppose the true hard region attached to the declared discovery event is constant, $\mathcal{H}_j = \mathcal{H}$, for a window of W comparison opportunities and each opportunity has the same deterministic compute cost $c > 0$. On every no-hit history in that window, suppose a memory policy M has conditional hit hazard at least p_M, and a comparator G has conditional hit hazard at most p_G, with $p_M > p_G$. Then, for $1 \le n \le W$,*

$$\mathbb{P}(\tau_M \le n) \ge 1 - (1 - p_M)^n > 1 - (1 - p_G)^n \ge \mathbb{P}(\tau_G \le n),$$

and, for $2 \le n \le W$,

$$\mathbb{E}[c \min(\tau_M, n)] \le c \sum_{j=0}^{n-1} (1 - p_M)^j < c \sum_{j=0}^{n-1} (1 - p_G)^j \le \mathbb{E}[c \min(\tau_G, n)].$$

For a generic policy that actually samples a parent, one sufficient derivation of p_M is the following: on each no-hit history, with conditional probability at least r a useful retained parent is available; given that event it is selected with probability at least γ; conditional on that selection, its mutation/evaluation reaches the true current hard region with probability at least α. Then $p_M \ge r\gamma\alpha$. No independence among these events is assumed; the bounds are conditional in the stated order.

That random-selection factorization is not the S6-AE schedule. For the verified deterministic Cartesian batch, every retained parent receives all 11 operators. If r lower-bounds availability of a useful parent and α_{batch} lower-bounds, under the explicitly declared RNG ensemble, the conditional probability that its scheduled operator batch generates the declared hit, then the appropriate sufficient bound is $p_M \ge r\alpha_{\text{batch}}$. With roots and all seeds conditioned to their frozen values, this is a statement about a realized batch, not an estimated random parent/operator-selection probability.

Importance. It turns a structural-memory claim into a finite-budget, equal-cost comparison that explicitly accounts for parent access and mutation reachability. It also makes clear that survival alone is not a sufficient mechanism.

Assumptions and realism. A piecewise-stable target window and equal-cost opportunities are useful first approximations. In S6-AE the BQ checkpoint is fixed during mutation/acquisition in cycle k and changes only after T_k is frozen and labeled, but comparison windows spanning adaptations can still move $\mathcal{H}_k$. HGB40, archive, exact-label, and BQ-training costs also differ from a single mutation attempt. The deterministic 10,000-by-11 schedule and $P = T$ update are verified; the history-uniform $r\alpha_{\text{batch}}$ bound, scientific threshold, and matched comparator cost/hazard remain [**Implementation applicability: OPEN**].

Proof approach. Induction on the no-hit survival probability, then the identity expressing expected truncated hitting time as the sum of survival probabilities. It requires no conditional independence and no Markov approximation. Proof details are in `proof_attempts.tex`.

Counterexample and HAC distinction. If memory's conditional hazard is lower because its retained parents are stale, or if a gradient proposal has larger hazard per compute unit, the premise fails. The exact examples in `counterexamples.tex` show both. Thus this is a conditional criterion that can distinguish BLADE from fixed generation or HAC only after their hazards and costs are measured; it does not establish that BLADE wins.

4.2 Candidate 2: inherited memory never hurts

[**Mathematical status: DISPROVED**] [**Implementation applicability: NOT APPLICABLE**] [**Scientific novelty: NOT CLAIMED**] The proposed blanket claim is that retaining historical parents cannot increase first-hitting cost relative to a fixed generator. This is false even with two structural states and one target change. The stale-memory example in Section 6.1 has expected cost 100 for inherited memory versus 2 for a fixed uniform generator. The failure is not a technical gap in Candidate 1: the required hazard dominance is reversed.

Importance and proof approach. A universal no-harm theorem would have supported the research objective directly, but the exact counterexample rules it out. No proof is possible without additional target-drift, forgetting, or reachability assumptions. These extra assumptions would need to be stated and independently justified; they cannot be inferred from the word "memory."

Assumptions and realism. The universal claim would require memory never to reduce true-hardness reachability per unit cost for any target shift or parent-selection history. That is not guaranteed by inherited populations and is contradicted by the stale-memory example; realism for BLADE-NCO is low absent explicit target-aware forgetting and verified hazard bounds.

Strongest proof approach. An exact finite-state counterexample proves the universal claim false. A positive restricted result would need additional drift and reachability premises, which are instead stated explicitly in Candidate 1.

HAC distinction. Nothing in this false claim compares BLADE to HAC. HAC can outperform a poorly aligned retained population, as the path example in Section 6.1 demonstrates.

4.3 Candidate 3: diversity always improves discovery

[**Mathematical status: DISPROVED**] [**Implementation applicability: NOT APPLICABLE**] [**Scientific novelty: NOT CLAIMED**] The proposed blanket claim is that preserving a more diverse population always reduces first-hitting time. In a three-state model, a diverse parent selector spends half of each equal-cost proposal on an irrelevant niche and has expected hitting cost 2, while a focused selector hits in cost 1. The exact construction is in Section 6.1.

Importance and assumptions. Diversity can preserve useful future routes, but its benefit depends on niche relevance, selection allocation, target drift, and cost. A monotonicity proof from

diversity cardinality alone cannot work. This counterclaim concerns a universal guarantee, not a claim that Elite-QD1000 behaves like the toy selector.

Realism and strongest proof approach. No selection-allocation or niche-relevance condition is supplied for Elite-QD1000, so the premise is not verified for BLADE-NCO. The exact three-state construction is a complete disproof of the universal claim; a positive diversity theorem would require explicit target-access conditions. This does not rank BLADE against HAC.

HAC distinction. It provides no general ranking against HAC. With equal compute, a focused or gradient-guided policy may spend more proposals on the current reachable hard region; the actual ranking depends on transition probabilities and costs, neither of which is specified here.

5 Proof attempt for the strongest candidate

5.1 History-based survival lemma

[Mathematical status: PROVED] [Implementation applicability: PARTIAL] [Scientific novelty: NOT CLAIMED] The lemma below is a standard finite-horizon hitting bound, included as the proof engine rather than as a novelty claim.

Let Y_j be the indicator of the declared discovery event at comparison opportunity j and let $\tau = \inf\{j \geq 1 : Y_j = 1\}$. The event $\{\tau \geq j\}$ is determined by the history before opportunity j. Assume there are deterministic numbers $p_j \in [0, 1]$ such that, on every pre-opportunity history with no previous hit,

$$\mathbb{P}(Y_j = 1 \mid \mathcal{F}_{j-1}) \geq p_j.$$

This is a history-uniform conditional hazard condition. It allows the target attached to the event to vary with j. The index j is deliberately not the BQ cycle index k: it may represent a serialized mutation attempt, a full generation event D_k^{gen}, or an acquisition event D_k^{acq}, provided the choice and filtration are fixed.

Lemma 1 (Non-stationary survival and cost bound)**.** *For every $n \geq 1$,*

$$\mathbb{P}(\tau > n) \leq \prod_{i=1}^{n} (1 - p_i).$$

If opportunity costs c_i are deterministic, then

$$\mathbb{E}[C_\tau] \leq \sum_{i \geq 1} c_i \prod_{j=1}^{i-1} (1 - p_j),$$

with the inequality valid also when the right side is infinite. For a budget B, assume $\sum_{i=1}^{\infty} c_i = \infty$ and let $n(B) = \max\{n \geq 0 : \sum_{i=1}^{n} c_i \leq B\}$, with empty sum zero.

$$\mathbb{P}(C_\tau \leq B) \geq 1 - \prod_{i=1}^{n(B)} (1 - p_i).$$

Proof. Write $S_i = \{\tau \geq i\}$ for survival to the start of opportunity i. Since $S_i \in \mathcal{F}_{i-1}$,

$$\begin{aligned}\mathbb{P}(S_{i+1}) &= \mathbb{E}[\mathbf{1}_{S_i}\mathbb{P}(Y_i = 0 \mid \mathcal{F}_{i-1})] \\ &\leq (1 - p_i)\mathbb{P}(S_i).\end{aligned}$$

Starting from $\mathbb{P}(S_1) = 1$ and iterating gives $\mathbb{P}(S_{n+1}) \leq \prod_{i=1}^{n}(1 - p_i)$, which is the first claim. For the cost, use the nonnegative-sum identity

$$C_\tau = \sum_{i \geq 1} c_i \mathbf{1}_{\{\tau \geq i\}}, \qquad \mathbb{E}[C_\tau] = \sum_{i \geq 1} c_i \mathbb{P}(S_i),$$

and apply the survival bound termwise. Finally, completion of any of the first $n(B)$ opportunities costs at most B, so budget success is at least the probability of a hit among those opportunities. The first claim supplies the stated lower bound. □

5.2 Deriving the window comparison

For Candidate 1, apply the survival lemma on the W-opportunity stable-target window. The memory policy satisfies $p_j = p_M$ and the comparator satisfies an upper hazard bound $q_j^G \leq p_G$ on every no-hit history. The same induction, with the inequality reversed for G, yields

$$\mathbb{P}(\tau_M > j) \leq (1 - p_M)^j, \qquad \mathbb{P}(\tau_G > j) \geq (1 - p_G)^j.$$

Therefore, for $n \leq W$,

$$\mathbb{P}(\tau_M \leq n) \geq 1 - (1 - p_M)^n > 1 - (1 - p_G)^n \geq \mathbb{P}(\tau_G \leq n).$$

For $2 \leq n \leq W$, using $\mathbb{E}[\min(\tau, n)] = \sum_{j=0}^{n-1} \mathbb{P}(\tau > j)$ and multiplying by the common opportunity cost c gives the strict truncated compute-cost comparison in Proposition 1. At $n = 1$ both policies pay the first opportunity cost in the truncated-cost convention, so only the success-probability comparison is strict there.

5.3 Dependency audit

- **Conditional independence:** not used. The proof requires history-conditional lower/upper hazard bounds; dependence across offspring or generations is allowed if those bounds remain true after every no-hit history.
- **Markovianity:** not used in the hitting proof. It would only be needed if one chooses a finite-state transition matrix to calculate the hazards. A niche aggregation is valid for that purpose only under lumpability.
- **Mutation bounds:** the quantity α must be a lower bound on reaching the true current hard region conditional on the useful parent being selected. A graph path's existence alone is insufficient; its probability and compute cost matter.
- **Parent retention and scheduling:** a lower bound r must concern availability of a genuinely useful parent at the declared opportunity. Historical presence before a target shift does not establish continued usefulness afterward. The S6-AE parent/operator schedule is deterministic: conditional on P_k, each parent receives all 11 operators. Hence no random parent-selection factor γ may be inserted; a batch reachability bound α_{batch} must be derived under the declared RNG ensemble.

- **Surrogate accuracy:** no accuracy is needed for the abstract survival lemma. To pass from D_k^{gen} to D_k^{acq}, a retention condition connecting $\widehat{h}_k$ and the QD rule to true h_k^{BQ} and future reachability is essential. Perfect current-hardness prediction alone is insufficient when tied candidates have different mutation futures. Without such a bridge, the implementation-level step is OPEN.
- **Cost normalization:** the comparison assumes equal deterministic cost c per opportunity. If BLADE archive updates, HGB40 inference, or BQ validation change cost, opportunity-count dominance does not imply compute-cost dominance.

5.4 Failure boundary

[**Mathematical status: OPEN for a BLADE-specific dominance claim**] [**Implementation applicability: PARTIAL**] [**Scientific novelty: OPEN**] The proof does not establish that BLADE's conditional hazard is above the fixed-generator or HAC hazard. It fails to produce a BLADE result at the exact step where one would need verified bounds on r and α_{batch} (or a direct generation/acquisition hazard bound) and a comparator upper bound p_G, all in the same compute units. The counterexamples show these inequalities can reverse. No unproved transition from a structural-memory narrative to hazard dominance is made.

6 Adversarial finite-state checks

The examples below are exact toy models. They test universal statements and assumptions; none is asserted to be the BLADE-NCO implementation. Their tags report mathematical status; implementation applicability is [**Implementation applicability: NOT APPLICABLE as a universal guarantee**], and novelty is [**Scientific novelty: NOT CLAIMED**].

6.1 Stale memory after a target shift

[**Mathematical status: DISPROVED**] for the claim that historical memory cannot hurt. Take $\mathcal{X} = \{a, b\}$. Before a change the hard region is $\mathcal{H}_0 = \{a\}$; just before the first measured proposal it shifts to $\mathcal{H}_i = \{b\}$ for all $i \geq 1$. The inherited population contains only a. The memory policy mutates from a to a with probability 0.99 and to b with probability 0.01, retaining a after a miss. Each proposal costs one unit. Its hitting time is geometric with success probability 0.01, so $\mathbb{E}[\tau_M] = 100$ and

$$\mathbb{P}(\tau_M \leq B) = 1 - 0.99^B.$$

A fixed generator that independently samples a or b uniformly every proposal has $\mathbb{E}[\tau_G] = 2$ and success probability $1 - 0.5^B$. Thus the memory policy is strictly worse at every positive integer budget. The target has undergone a single non-stationary shift; the failure is stale structural support, not an independence artifact.

6.2 Gradient adaptation can beat structural mutation

[**Mathematical status: DISPROVED**] for any universal claim that structural mutation outperforms HAC-style gradient adaptation. Let $\mathcal{X} = \{0, 1, \ldots, L\}$ and $\mathcal{H} = \{L\}$. A structural mutation

from $i < L$ advances to $i+1$ with probability ε and otherwise stays at i. Starting at 0, its expected number of unit-cost proposals to hit L is L/ε. A gradient-informed comparator that advances deterministically one step per unit-cost proposal hits in L proposals. For any $0 < \varepsilon < 1$, the structural policy is slower. This is a toy landscape with an informative gradient, not a model of the actual HAC implementation; it proves that operator labels alone imply no ranking.

6.3 Diversity can allocate effort away from the target

[**Mathematical status: DISPROVED**] for the claim that increased diversity always improves first hitting time. Let $\mathcal{X} = \{s, u, g\}$, target $\mathcal{H} = \{g\}$, and one proposal cost one unit. The diverse parent set contains distinct niches s and u; parent selection is uniform. Mutation from s reaches g surely, whereas mutation from u stays at u. Hence the per-proposal hit probability is $1/2$ and expected hitting cost is 2. A focused policy that selects s every time hits in cost 1. Preserving the irrelevant niche does not help this target and reduces access to the useful parent.

6.4 Perfect current-hardness prediction need not preserve a bridge

[**Mathematical status: DISPROVED**] for the claim that perfect prediction of current BQ hardness makes score-elite retention preserve future mutation reachability. Consider states $\{a, b, g\}$ and current target $\mathcal{H} = \{g\}$; neither a nor b is itself a hit, and their current true hardness is equal: $h(a) = h(b) < \kappa$. From a, the next scheduled mutation reaches g surely; from b, mutation stays at b surely. Let the surrogate be perfect on current hardness, so $\widehat{h}(a) = h(a) = h(b) = \widehat{h}(b)$. A one-slot score-elite update must still break the tie. If its deterministic tie rule retains b, it discards a and makes g unreachable. Thus current-hardness accuracy, even perfection, does not identify equal-hardness candidates with different mutation futures. A reachability-aware rule could retain a, but perfect current-hardness prediction alone need not do so. This is not a claim that the observed S6-AE hash tie-break selected such a pair; it is an exact obstruction to the proposed bridge.

6.5 Niche aggregation can destroy the Markov property

[**Mathematical status: DISPROVED**] for the assertion that a niche label alone is a Markov state. Let full states x_1, x_2 both map to niche A, while z, w map to distinct niches Z, W. From x_1, the next state is z with probability 1; from x_2, the next state is w with probability 1. Both full states have the same current aggregate A, but the next aggregate is respectively Z or W. The transition probability out of aggregate A is therefore not determined by the aggregate alone (strong lumpability fails). Therefore niche aggregation is not Markov uniformly over initial laws. For a specially chosen initial distribution, an averaged transition might nevertheless be Markov; that weaker distribution-dependent claim would require the conditional within-A mixture to remain history-independent. The example does not disprove every such weak aggregation. A proof using only niche counts is invalid unless it proves the relevant strong or distribution-dependent condition, or augments the state with the hidden structural/solver information.

7 Research assessment

Strongest candidate. The useful candidate is the finite-window reachability comparison in Proposition 1. It gives an auditable sufficient condition for inherited memory to improve finite-budget discovery on a window where the hard region is stable: the retained lineage must be available, selected, and able to mutate into the *true* current hard region often enough that its per-cost conditional hit hazard exceeds the comparator's. The proof is a survival-probability induction and needs neither independent proposals nor a Markov approximation.

Limitations and novelty. The probability argument is standard and is not itself a novel theorem. Its scientific content would have to come from deriving and validating useful lower bounds on FOGA-11 reachability and parent survival, then comparing them with the actual fixed-generator and HAC hazards at equal compute cost. Stage 1.5 verifies the deterministic operator schedule and population update, but the operator-kernel laws under a scientific ensemble, hardness threshold, score-to-retention bridge, matched costs, and HAC transition law remain unspecified. Thus the decisive assumptions cannot yet be checked, and the result does not assert a BLADE advantage.

Stage 1.5 verified the S6-AE batch schedule and population update rule, but not the required probabilistic hazard bounds, a scientific hard-region threshold, or a matched HAC transition/cost law. Therefore the mathematical comparison remains conditional, its BLADE applicability remains partial, and its novelty remains open.

Testability. The result could be tested from existing proposal- and generation-level records if they identify parent, operator, accepted child, HGB40 score, acquisition membership, current BQ hardness where evaluated, solver checkpoint, survival, and comparable cost. The inspected S6-AE artifacts do record much of this mechanism, but they do not by themselves establish the history-uniform true-hardness hazard bounds used by the proposition.

[Implementation applicability: OPEN]

No new evaluations, training, or evolutionary runs were performed for this note.

Unresolved issues. The decisive open issue is whether memory yields greater *true-hardness reachability per unit cost* than the fixed generator or HAC after accounting for stale parents, surrogate ranking error, solver co-adaptation, and archive overhead. The exact BLADE transition model and the appropriate cost accounting must be supplied before a BLADE-specific theorem can be attempted.

8 Stopping point

Stage 1 formalization, the Stage 1.5 corrections, candidate claims, original proof attempt, and adversarial checks are recorded. No Stage 2 theorem development, experimental design, algorithm change, Oracle/BQ evaluation, or new run is included.

References

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[2] P. S. Oliveto and C. Zarges, "Analysis of diversity mechanisms for optimisation in dynamic environments with low frequencies of change," *Theoretical Computer Science*, vol. 561, part A, pp. 37–56, 2015. doi: 10.1016/j.tcs.2014.10.028.